



Food and Agriculture
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Science for resilient livelihoods in dry areas



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WEBINARS SERIES ON RAINWATER HARVESTING

25th October 2022, 13 h 00 – 14 h 30 (GMT+2)

Module n°8:

**Rainwater harvesting systems for
smallholders**



Session: Socio-economic benefits and risks of RWH – Livelihood transformation

Date: Tuesday 25th October 2022 **Time:** 13 h 00 – 14 h 30 (GMT+2)

Register: [Socio-economic benefits and risks of RWH – Livelihood transformation](#)



BACKGROUND

Water scarcity is a critical challenge in countries and regions with arid and semi-arid conditions and fast-growing water demands. This is mostly the case in all of the countries across the NENA Region, where food demand and food insecurity continue to grow, widening the water supply-demand gap in most countries. Furthermore, the anticipated impact of climate change in the NENA region is indicative towards an increased intensity of rainstorms and flash floods, while the total amount of annual precipitation rates shows declining trends. During the last 20 years, interest has been renewed in rainwater harvesting (RWH) as one of the most effective climate adaptation strategies to cope with water scarcity, thus reducing the pressure on the limited freshwater resources and increasing rainfed crop productivity.

This webinar series on Rainwater Harvesting aims to provide an overview of RWH applications and illustrates, with country examples, the diversity of approaches. It will try to unlock the potential of rainwater harvesting applications in the NENA region by highlighting the role of rainwater management in overcoming the current challenges in achieving water security. It will also shed the light on the need for innovations in economics, social consensus and supportive laws and regulations to ensure a widespread uptake of RWH technologies in the NENA region.

Main Conveners:

FAO/WEPS-NENA project & Water Scarcity Initiative (WSI) with: ICARDA, ICID and FAO inter-Regional Technical Platform on Water Scarcity (RTP-WS)

Contributing Partners:

The Arab Organization for Agricultural Development (AOAD) and United Nations Economic and Social Commission for Western Asia (UNESCWA)

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Languages:

English and French with live translation

Recording:

<https://dgroups.org/fao/waterproductivity/rwh>

ABOUT THE MODULE 8

This module will explore the role of rainwater harvesting systems in sustaining smallholders' livelihoods. It will highlight the role of rainwater harvesting on strengthening the livelihood security of rural communities and building their resilience to climate change. The discussions will draw from various case studies from the Arab region.



SUGGESTED READINGS

- ➔ Abu Hammad A., Borressen T. (2006) Socioeconomic Factors Affecting Farmers' Perceptions of Land Degradation and Stone wall Terraces in Central Palestine, *Environmental Management*, 37(3): 380-94.
- ➔ Ghanem M. et al. (2020) Socio-Economic and Environmental Impacts Assessment of Using Different Rainwater Harvesting Techniques in Sarida Catchment, West Bank, Palestine
- ➔ Jesmi, Y. et al. (2014). Risk Assessment of Rooftop-Collected Rainwater for Individual Household and Community Use in Central Kerala, India. *Journal of environmental health*. 76. 114-21.
https://www.researchgate.net/publication/260946473_Risk_Assessment_of_Rooftop-Collected_Rainwater_for_Individual_Household_and_Community_Use_in_Central_Kerala_India/link/odeec53c626eb85b55000000/download
- ➔ Mutekwa V. and Kusangya S. (2007) Contribution of rainwater harvesting technologies to rural livelihoods in Zimbabwe: The case of Ngundu ward in Chivi District. Department of Geography & Environmental Science, University of Zimbabwe
<https://www.semanticscholar.org/paper/Contribution-of-rainwater-harvesting-technologies-Mutekwa-Kusangya/a/dbb361f95c1e5c483d95cac99d42aed17e64a417>
- ➔ Omer, M. A. and Alamin E. M. (2011) Supplementary Irrigation and Soil Amendment Management with Sorghum on Khor Abu Habil (Sudan). *Journal of Environmental Science and Engineering*, David Publishing Company, 5(2011) 1079 – 1083, USA.
- ➔ Omer, M. A.; Saxton, K.E. and Bassett, D. L. (1988) Optimum sorghum planting dates in western Sudan by simulated water budgets. *Agric. Water Management*, 13 (1988):33 – 48. Elsevier Science Publisher H. V., Amsterdam, The Netherlands.
- ➔ Omer, M. A. and Elamin, E. M. (1996) Effect of tillage and contour diking on sorghum establishment and yield on sandy clay soil in Sudan. *Soil and Tillage Research*: 43 (1997) 229-240. Elsevier Science Publisher H. V., Amsterdam, The Netherlands.
- ➔ Tamimi L. (2016) Rainwater Harvesting System. Quality And Impacts On Public Health, Faculty of Graduate Studies, Birzeit University



CERTIFICATE OF ACHIEVEMENT

A digital certificate of achievement will be offered by FAO to participants who:

- Attend at least four webinars belonging to Module 1 to Module 4 and five webinars belonging to Module 5 to Module 10;
- Obtain a score of no less than 70 percent in each of two written assessments that include multiple-answer questions and reflect the content of the aforementioned modules.

The first assessment covering the modules 1 to 4 was organized in August 2022. The link to the second assessment, covering the modules 5 to 10, will be sent by email at the end of the year to eligible participants who will be given two weeks to submit their answers. Respondents will have the possibility to make two attempts to answer the assessment questions, and only the highest score will be considered.

Important note: For participants who wish to obtain a certificate of achievement, to be able to track your attendance, you must register and join the webinars using the same e-mail address.

Speakers and moderator



Wael Abu Rmaileh

received a B.Sc. in Agricultural Engineering-Soil and Irrigation Department (University of Jordan, 1992), and earned MSc degree in Natural Resources & Sustainable Management (Al-Khalil University - Palestine, 2009). He started working at the Land Research Center (LRC) 25 years ago, focusing on the planning, financing and implementation of several agricultural, environmental and soil, water projects. Wael Abu Rmaileh has gained an extensive experience through implementing many projects in the West Bank - Palestine related to Water Harvesting Techniques of many watersheds, climate and environment justice, food security, Land Degradation Assessment, Land Suitability for Development. All these activities were carried out in cooperation with the Ministry of Agriculture and local and international institutions.



Mekki Abdellatif Omer

holds a B.Sc. (Honor) Agric. Engineering (Khartoum University, Sudan), M.Sc. Engineering Science (Irrigation) and Ph.D. (Hydrology) from Washington State University (USA). He served as a Research Scientist at the Agricultural Research Corporation (1987-2020) with a focus on agricultural water management. His profession also includes large working experience and Consultant services to many national, regional, and international agencies such as FAO, IFAD, UNDP, AOAD, IDA, CARE International, BADEA for the development, design, implementation, monitoring and evaluation of development projects and capacity building in Sudan and various other countries (Yemen, Syria, KSA, Somali, etc.). He is currently Supervisor of the Agricultural, Environmental and Climate Change Programs Unit at AOAD in Sudan.



Hichem Charieg

is Programme Monitoring and Reporting Expert at the FAO Regional Office for the Near East and North Africa, member of the Regional Initiative on Water Scarcity. Hichem Charieg has a background in agricultural economics engineering and holds a master degree in Economy of Agriculture, Agribusiness and Environment with focus on natural resources management. He has worked on planning and implementing several development projects with the private sector, NGOs, research centers, and UN agencies in the Arab region and West African countries. During his work with FAO, he has been involved in different water related activities in relation with planning, reporting, partnership, data production, knowledge materials, etc.

Module 8: Rainwater harvesting systems for smallholders

Session: Socio-economic benefits and risks of RWH – Livelihood transformation

Date: Tuesday, 25th October, 2022

Time: 01:00 PM - 02:30 PM (GMT +2)

Time	Activity
01:00 – 01:05	Welcoming Remarks Introduction to the session objectives and speakers By: Hichem Charieg , Programme Monitoring and Reporting Expert at the FAO Regional Office for the Near East and North Africa
01:05 – 01:30	Presentation on “Rainwater harvesting and rural communities, strengthened livelihoods in Palestine” By: Wael Abu Rmaileh , ESCWA Consultant on rainwater harvesting systems
01:30 – 01:55	Presentation on “The role of RWH in building capacity of traditional rain-fed farmers and strengthening livelihood security in Darfur” By: Mekki Abdellatif Omer , Supervisor of the Agricultural, Environmental and Climate Change Program Unit, Arab Organization for Agricultural Development, Sudan
01:55 – 02:25	Open Discussions – Q&A's
02:25 – 02:30	Conclusion & Wrap-up



This activity is implemented within the regional project "Implementing the 2030 Agenda for water efficiency/productivity and water sustainability in NENA countries" under the umbrella of the Water Scarcity Initiative. This regional project is implemented by the Food and Agriculture Organization of the United Nations and funded by the Swedish International Development Cooperation Agency (SIDA).

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